

FloodRiskAI Platform Presentation

Next-Generation AI Disaster Intelligence & Early Warning System

DataQuest 2026 · Disaster Management Analytics (Level 1 + Level 2 + Level 3)

Vignan's Foundation for Science, Technology & Research

Executive Summary & Problem Statement

The Disaster Challenge: Annual monsoon flooding affects millions across Indian river basins (Assam, Kerala, Bihar, Odisha, Maharashtra). Existing tools lack integrated AI risk scoring, real-time river telemetry, climate stress simulation, and automated emergency playbook dispatch.

Core Objectives	Platform Innovation
• Transition from static historical reporting to real-time predictive AI intelligence. • Provide local SHAP explainability for every risk assessment. • Automate Disaster Response Playbooks for District Disaster Management Authorities (DDMA).	• 3-Tiered Platform: Level 1 Analytics + Level 2 Multi-Model AI + Level 3 Climate Simulation. • Location Search Engine: Instant flood risk prediction for 50+ Indian cities. • Interactive GIS Mapping: River telemetry stream + state heatmaps.

System Architecture: 3-Tiered Intelligence Platform

LEVEL 1 · ANALYTICS	LEVEL 2 · AI PREDICTION	LEVEL 3 · INTELLIGENCE
• Historical timeline trends (1967-2023) • Severity class ratio analysis • Primary meteorological trigger breakdown • Real-time river telemetry monitoring stream	• Multi-Model Suite (Random Forest, SVM, Logistic Regression) • Local SHAP feature waterfall attributions • Animated SVG circular risk gauge • Historical event quick-loaders	• Climate scenario stress simulator • Peak discharge sensitivity curves • Location search geocoding engine • Automated DDMA response playbook dispatch

Data Infrastructure & Benchmark Datasets

Gauge Flood Events Dataset (`floodevents_indofloods.csv`)	4,548 Events Indo-Floods Benchmark	Peak Discharge (m^3/s), Flood Volume, Event Duration, Time to Peak, Recession Time, Start Month, Flood Type (Moderate vs. Severe). Used for ML model training.
India Flood Inventory Dataset (`India_Flood_Inventory_v3.csv`)	6,876 Records IMD & NDMA Historical	State & District boundaries, Main Causes (Heavy Rain, Cyclone, Cloudburst), Fatalities, Displaced Persons, Area Affected. Used for Level 1 GIS Heatmaps.
River Telemetry Stream (Live JSON Stream)	Major River Basins Brahmaputra, Ganga, etc.	Real-time river stage levels vs. danger thresholds, discharge rates, telemetry trend (rising/receding), and status tags (CRITICAL, WARNING, NORMAL).

Machine Learning Suite & Performance Benchmarks

Evaluated multiple machine learning algorithms on test data split (25% holdout) with standard scaling:

Model Architecture	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC Score
Random Forest Classifier (Primary)	82.5%	75.4%	75.9%	75.6%	0.879
Support Vector Machine (SVM)	71.3%	77.9%	27.8%	40.9%	0.763
Logistic Regression Classifier	69.8%	62.1%	34.5%	44.3%	0.742

SHAP Feature Contribution Decomposition: Local feature impacts compute how parameters like *Peak Discharge* (+18.5%) and *Flood Volume* (+12.1%) push risk scores up relative to population baseline means.

Location Intelligence & Place Risk Search

Users can search **any Indian City, District, or State** (e.g. *Guwahati, Wayanad, Patna, Kochi, Cuttack, Mumbai, Delhi, Varanasi*) to instantly predict location-specific flood risk.

Guwahati, Assam	Brahmaputra Basin	14,200 cumec	82.0% (HIGH RISK)
Wayanad, Kerala	Kabini / Chaliyar Rivers	7,200 cumec	63.5% (MEDIUM RISK)
Patna, Bihar	Ganga Basin	8,900 cumec	74.2% (HIGH RISK)
Mumbai, Maharashtra	Mithi / Coastal Drainage	5,200 cumec	52.1% (MEDIUM RISK)

Emergency Alert Center & Automated Response Playbooks

Automated threat level classification triggers instant localized operational action plans for District Disaster Management Authorities (DDMA):

RED ALERT (CRITICAL)	Water Level > Danger Level Risk Score ≥ 66%	• Mandatory floodplain evacuation within 2km radius. • Dispatch 2 NDRF search-and-rescue teams with motorboats. • Ready 12 government school shelters (capacity 15,000 displaced persons).
YELLOW ALERT (WARNING)	Water Level near Danger Risk Score 33% - 66%	• Issue preliminary warning advisories to low-lying villages. • Standby SDRF units alerted. • Increase telemetry monitoring frequency to hourly interval readings.
NORMAL	Water Level Normal Risk Score < 33%	• Routine gauge recording and standard operational telemetry data logging.

Deployment Infrastructure & Live Working Links

Deployment Target	URL / Location	Description
Localhost Web Server	<code>`http://127.0.0.1:5000`</code>	Flask WSGI Server with full Glassmorphism UI, GIS Leaflet maps, and live API endpoints.
GitHub Repository	<code>`https://github.com/darshitha02/floodriskmanagement`</code>	Main source repository with commit history, ML models, and deployment configurations.
Render Cloud Web App	<code>`https://floodriskmanagement.onrender.com`</code>	Production Flask web server deployment via Gunicorn (<code>`Procfile`</code> & <code>`render.yaml`</code>).
Streamlit Cloud App	<code>`https://share.streamlit.io`</code>	Native Streamlit interactive cloud app (<code>`streamlit_app.py`</code>) with Plotly & Folium integration.

Conclusion: FloodRiskAI delivers an extraordinary end-to-end disaster intelligence system empowering disaster management authorities, researchers, and citizens with accurate, explainable, and actionable flood risk insights.